Testing Strategy

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 04_Transition/Testing_Strategy.md

Full testing strategy for Resilio covering the testing methodology, test levels, test design approach, and the rationale for the test coverage achieved. This document supports Chapter 4 of the FYP report.

TESTING OVERVIEW

Resilio was tested across three levels following a bottom-up strategy: unit tests validated individual components in isolation, system tests validated integrated end-to-end flows, and user acceptance tests validated the system against real-world user requirements with external participants.

Test Level	Quantity	Pass Rate	Conducted By
Unit Tests	40 (30 documented + 10 in repo)	100%	Developer
System Tests	40 (15 documented + 25 in repo)	100%	Developer
User Acceptance Tests	25	104/105 (99.05%)	External participants
Total	**105**	**99.05%**	—

TESTING METHODOLOGY

Approach: Black-Box and White-Box Hybrid

Type	Applied To	Description
White-box testing	Unit tests (TC-U series)	Internal logic tested directly: database constraints, validation rules, RLS policies, business logic in service layer

Type	Applied To	Description
Black-box testing	System tests (TC-S series) + UAT (TC-UAT series)	System tested through its public interfaces: API endpoints and Flutter UI; no assumption about internal implementation

Technique: Equivalence Partitioning and Boundary Value Analysis

Applied to input validation tests:

- **TC-U009:** Mood score = 0 (below valid minimum of 1) → boundary value
- **TC-U010:** Mood score = 6 (above valid maximum of 5) → boundary value
- **TC-U012:** PHQ-9 answer = 4 (above valid maximum of 3) → boundary value
- **TC-S006:** PHQ-9 total = 20 (crisis threshold boundary) → boundary value

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TEST LEVEL 1: UNIT TESTS (TC-U001 to TC-U040)

Scope

Unit tests target individual system components:

- Authentication service (SuperTokens integration)
- Appointment repository (creation, conflict detection, status updates)
- Mood entry service (validation, persistence)
- Questionnaire service (score calculation, severity band assignment, crisis threshold)
- Payment verification service (eSewa callback processing, tamper detection)
- Subscription service (creation, duplicate prevention)
- Game session service (recording, achievement trigger)
- Favourites service (creation, duplicate prevention)
- Therapist profile service (linking, verification status)
- Notification service (creation on events)
- Row Level Security policies (cross-user access prevention)
- Offline sync (SQLite write → remote sync)

Test Environment

- **Database:** Supabase test project (separate from production)
- **Authentication:** SuperTokens test tenant
- **Payment:** eSewa test environment

• **Notifications:** FCM test token (non-production)

Coverage by Module

Module	Test Cases	All Pass
Authentication (OTP + Social)	TC-U001 to TC-U004	Yes
Appointments	TC-U005 to TC-U007	Yes
Mood Entries	TC-U008 to TC-U010	Yes
Questionnaires (PHQ-9, GAD-7)	TC-U011 to TC-U013	Yes
Payment (eSewa)	TC-U014 to TC-U016	Yes
Subscriptions	TC-U017 to TC-U018	Yes
Games and Achievements	TC-U019 to TC-U020	Yes
Favourites	TC-U021 to TC-U022	Yes
Therapist Profile	TC-U023 to TC-U025	Yes
Notifications	TC-U026	Yes
Row Level Security	TC-U027 to TC-U028	Yes
Offline Sync	TC-U029 to TC-U030	Yes
Advanced (in repo)	TC-U031 to TC-U040	Yes

TEST LEVEL 2: SYSTEM TESTS (TC-S001 to TC-S040)

Scope

System tests validate complete end-to-end user flows across all three modules, ensuring the Flutter frontend, Node.js API, Supabase database, and external services (eSewa, FCM, WebRTC, Cloudinary) work together correctly.

Key Flows Tested

Flow	Test Cases
Complete customer registration and onboarding	TC-S001

Flow	Test Cases
Therapist discovery, filtering, and profile view	TC-S002
End-to-end appointment booking with eSewa payment	TC-S003
Live WebRTC video consultation	TC-S004
Therapist access to patient data during session	TC-S005
PHQ-9 severe score crisis protocol	TC-S006
Therapist content publication through admin approval	TC-S007
Admin therapist registration rejection	TC-S008
Wellness game session recording	TC-S009
Achievement badge display after unlock	TC-S010
Push notification delivery on appointment confirmation	TC-S011
Subscription-gated premium content access	TC-S012, TC-S013
Mood chart rendering	TC-S014
Audio track playback from content hub	TC-S015
Additional regression tests (in repo)	TC-S016 to TC-S040

System Test Environment

- **Device:** Physical Android device (Pixel-class hardware) + second device for WebRTC sessions
- **Network:** Both WiFi (for WebRTC quality testing) and 4G mobile (for TURN server validation)
- **Backend:** Development server with production-equivalent configuration
- **Database:** Staging Supabase project (separate from UAT data)

TEST LEVEL 3: USER ACCEPTANCE TESTING (TC-UAT001 to TC-UAT025)

UAT Objectives

Validate that real users (not the developer) can complete all primary user journeys without assistance

Validate that external clients (Ms. Karki, Dr. Singh) confirm the platform meets the requirements they specified

Identify any usability issues not surfaced by functional testing

Measure user satisfaction and willingness to recommend

UAT Participants

Participant	Type	Role in UAT
Female, age 24	Target user (student)	Customer journey: registration → booking → mood → content
Male, age 29	Target user (working professional)	Customer journey: content hub → subscription → games
Male, age 26	Target user (young professional)	Offline testing: airplane mode mood logging
Male, age 31	Target user (professional)	Notification testing: push notification receipt and deep link
Female, age 22	Target user (student)	Therapist discovery + trust rating
Ms. Sushana Karki	External client	Full therapist role: content creation, earnings, session workflow
Dr. Dibyandra Singh	External client	Admin role: therapist verification, content moderation, analytics

Note: Only the 5 target users completed TC-UAT001 to TC-UAT013 and TC-UAT021 to TC-UAT025. Ms. Karki completed TC-UAT014 to TC-UAT016 and TC-UAT020. Dr. Singh completed TC-UAT017 to TC-UAT019.

UAT Protocol

Each target user session followed this protocol:

Briefing (5 min): Participant told the purpose of the session; no instructions given about how to use the app

Task completion (45–60 min): Participant works through assigned tasks unassisted; observer notes any confusion or assistance requests

Feedback form (10 min): Participant completes the written feedback form (Appendix L)

Debrief (5–10 min): Participant asked to expand verbally on any feedback they gave

Assistance rule: The test facilitator (developer) did not provide any guidance during the task completion phase. If a participant became completely stuck, this was recorded as a failure or partial result for that test case.

UAT Result: TC-UAT011 (Partial Pass)

TC-UAT011 tested whether a user could locate the PHQ-9 questionnaire within the app without instruction. The participant found the questionnaire after approximately 3 minutes — exceeding the 2-minute expected threshold. No assistance was required, so the test was recorded as **PARTIAL** rather than FAIL. The navigation path to the questionnaire section has been noted as a UX improvement for the next release (more prominent placement in bottom navigation or home screen quick action).

TEST TRACEABILITY MATRIX

This matrix maps each functional requirement from the SRS to the test cases that validate it.

Requirement	Test Cases
FR-C01: OTP/Google/Facebook registration	TC-U001 to TC-U003, TC-S001
FR-C02: Preference onboarding	TC-S001, TC-UAT002
FR-C03: Therapist directory browsing	TC-U024, TC-S002, TC-UAT003
FR-C04: Matching questionnaire	TC-UAT003
FR-C05: Appointment booking	TC-U005 to TC-U007, TC-S003, TC-UAT004
FR-C06: eSewa payment	TC-U014 to TC-U016, TC-S003, TC-UAT004
FR-C07: WebRTC consultation	TC-S004, TC-S005, TC-UAT012
FR-C08: Mood logging	TC-U008 to TC-U010, TC-S014, TC-UAT005, TC-UAT006
FR-C09: PHQ-9 and GAD-7	TC-U011 to TC-U013, TC-S006, TC-UAT007, TC-UAT008
FR-C10: Content hub	TC-S012, TC-S013, TC-S015, TC-UAT010
FR-C11: Wellness games + achievements	TC-U019, TC-U020, TC-S009, TC-S010, TC-UAT009
FR-C12: Push notifications	TC-U026, TC-S011, TC-UAT016, TC-UAT024
FR-C13: Favourites	TC-U021, TC-U022
FR-T01: Therapist registration	TC-U023, TC-UAT017
FR-T02: Availability calendar	TC-U005, TC-UAT014
FR-T03: Appointment list + patient records	TC-S005, TC-UAT020
FR-T04: Patient data in consultation	TC-S005
FR-T05: Content publishing	TC-S007, TC-UAT014

Requirement	Test Cases
FR-T06: Earnings summary	TC-UAT015
FR-A01: Therapist verification	TC-U024, TC-U025, TC-S008, TC-UAT017
FR-A02: User management	TC-UAT017 (indirectly)
FR-A03: Content moderation	TC-S007, TC-S008, TC-UAT018
FR-A04: Platform analytics	TC-UAT019

DEFECTS LOG

ID	Test Case	Defect Description	Severity	Resolution
BUG-001	TC-U007	Double-booking check was not atomic — race condition possible under concurrent requests	High	Database-level unique constraint added to prevent race condition
BUG-002	TC-U016	eSewa tampered callback was initially accepted due to string comparison instead of numeric	High	Changed amount comparison to parseFloat with tolerance check
BUG-003	TC-S004	WebRTC connection failed on 4G mobile — STUN insufficient	High	TURN server deployed (R-T01 mitigation)
BUG-004	TC-UAT011	PHQ-9 questionnaire not discoverable within 2 minutes	Low	Navigation path noted for next release UX improvement
BUG-005	TC-S001	Google Sign-In redirect URL misconfigured in SuperTokens dashboard	Medium	Redirect URL corrected in SuperTokens dashboard configuration

Total defects found: 5

Critical/High defects resolved before submission: 3 of 3

Medium defects resolved: 1 of 1

Low defects (deferred): 1 (TC-UAT011 — navigation improvement)

TESTING TOOLS

Tool	Used For
Postman	API endpoint testing during unit and system testing
Android Studio Emulator	Functional testing during Construction iterations
Physical Android device	FCM, WebRTC, eSewa, offline mode testing
Flutter DevTools	Widget inspection; memory and performance profiling
Supabase Dashboard	Verifying database state after each test case
`flutter test`	Running automated widget and unit tests